

Architectural Specification

DOUBLE DECK PARKING SYSTEM

SECTION 11.2

Rev	Date	Description
0	14 APR 2022	First Issue

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DOUBLE DECK PARKING SYSTEM

PART 1 GENERAL

1.01 SUMMARY OF WORK

- A. The Contractor shall design, supply, fabricate, deliver, install, test, adjust, commission and maintain the works constituting the installation for the Double Deck Parking System as indicated on the drawings and this specification, or necessary works for the safe and proper operation of the whole system.
- B. All the ancillary equipment including sundry components that are necessary for the complete installation and proper operation of the Double Deck Parking system, including all labours, whether or not these equipment or sundries are specified are deemed to be including in the works.
- C. The said installation works are hereinafter named as Double Deck Parking System to the captioned building, all in conformity with the drawings, specifications, conditions of Contract and all relevant regulations including any measures that may be required to that end, notwithstanding any omissions or inadequacies of the drawings and specification. Without limiting the generalities of the foregoing, the Double Deck Parking System shall include, but not limited, to the following :-
 - 1. The system shall consist of Driving, Control & Safety system as stipulated in this specification and as shown on drawings.
 - 2. The mechanical parking system shall be designed and constructed according to the international standard such as GB, JIS, EN ... etc. the Double Deck Parking system shall be sufficient to withstand typhoon signal no. 10 and weatherproof.
 - 3. All equipment metal frame, pallet, anchors, attachments, reinforcement, and steel reinforcing for the system required for the complete installation.
 - 4. Supply and installation of 32A 1-Phase electric vehicle chargers mounted on steel frame of parking system.
 - 5. Anti-corrosive treatment on all metals used in the system.
 - 6. Control and other E&M requirements necessary for the safe operation of the system.
 - 7. Earthing Bonding for all exposed metallic parts.
 - 8. Undertaken letter for the mechanical car park system operation, which have to be provided by the supplier.
 - 9. Prepare necessary maintenance plan and contingency plan for the system, to avoid vehicular traffic queuing back on public carriageway.

10. The safety barrier to be provided along the side of parking machine face to driveway or passage.
11. A Registered Professional Engineer (RPE) of a suitable discipline (e.g. mechanical, marine etc.) should be engaged for undertaking examination on the double-deck car parks and certifying the equipment to be in safe working order on an annual basis.
12. The Contractor shall make all necessary arrangements and applications to the EMSD for the necessary testing, inspection and issuing of Certificate of operation by the Department for the whole installation and to the satisfaction of Transport Department, EMSD and Buildings Department.
13. The Contractor shall also be responsible for providing the following:
 - a. Double Deck Parking System and submission of all associated type of test certificate / prescribed forms for the operation of the Double Deck Parking system.
 - b. Shop Drawings: submit installation drawings showing location of each machine part, the details of machine construction, anchorage provisions and other components. Anchorage provision shall conform to the structural requirement of the building regulations.
 - c. Delivery schedule, programme and monitoring of the work.
 - d. Control wiring diagram
 - e. Mock-ups and test units.
 - f. On site testing of the Double Deck Parking System on the completion.
 - g. Coordination with work of other trades.
 - h. Protection of all materials and works.
 - i. Protection of existing property and warning sign to public.
 - j. Preparation of submission drawings and structural justification to Buildings Authority.
 - k. As-built record drawings, O&M Manual and photographs with DVD-rom before handover.
 - l. Guarantees and warranties.
 - m. Finish painting and colour of steel frame shall be submitted for the Architect's approval.
 - n. Relevant permanent notices and signage for the Double Deck Parking System to meet latest Building Regulations, FSD and EMSD requirements.
 - o. List of spare parts.
14. The structural stability of the whole System is the prime responsibility of the Contractor. Any defect found within the Guarantee Period shall be sealed and made good at the Contractor own expense.
15. All hoisting, staging, hoarding and temporary services including diversion of traffic shall be provided by the contractor.
16. Final cleaning of the system and removal of protection measures to the system.
17. The Contractor shall be responsible for verifying all the dimensions and actual conditions on site before manufacture.
18. Free comprehensive services and maintenance and emergency call service to the double-deck car parking system installation for a period of 12 months including all fee required by Government Authorities after the date of Substantial Completion issued by Architect.

19. The Contractor shall include the cost for third party structural justification reports and submission to Government Authorities, third party risk assessment reports and submission to Government Authorities and the equipment submission to laboratory testing if required by relevant government authorities.
20. A full time attendance (office hours) for statutory inspection & staff training for the first three months after Substantial completion certificate has been issued.
21. Customized training of the Employer's staff in operating and maintaining the system.
22. Provide qualified full time site engineer for coordination as required and approved by Architect.
23. Electrical Supply and Electrical Installation
 - a. The contractor shall provide power supply cable complete with an isolating switch adjacent to each set of double deck car parking mechanical system.
 - b. The contractor shall provide all electrical connections from the isolating switch, including the outgoing wiring either in trunking or conduits / the sub-main switches / distribution boards to the machines, control equipment, lighting and power, earthing bonding etc. for the mechanical parking system. All the electrical installation by mechanical parking Contractor shall be weatherproof type except those approved by architect / engineer.
 - c. Electricity supply shall be 380 volts/3 phase/50 Hertz or 220 volts/single phase/50 Hertz.
 - d. In the event of failure of electricity supply or fire breaks out, a mechanical device shall be provided to raise and lower the platform by the Contractor.
 - e. All exposed metal work liable to become electrically charged shall be efficiently bonded and earthed. All wiring shall be enclosed in galvanized steel conduits or trunking. Steel conduit, except flexible conduits shall be of heavy gauge and screwed.

1.02 QUALITY ASSURANCE

(NOT USED)

1.03 PERFORMANCE REQUIREMENTS

- A. Notwithstanding to any contrary contained herein pertaining to this installation, it shall be the Contractor's responsibility to ensure all the works is in strict accordance with the following statutory obligations, regulations and specifications together with any amendments made thereto :-
 1. The relevant British Standards and Codes of Practice Published by the British Standard Institution, London.
 2. Guide to Safety on Double-Deck Car Park of latest edition and amendments issued by the Electrical and Mechanical Services Department.
 3. The Regulations for Electrical Installation published by the Institution of Electrical Engineers, London
 4. The requirements of The Transport Department.
 5. General Specification for Building Edition by Architectural Services Department, Hong Kong.
 6. General Specification for Electrical Installation in Government Building, of the Hong Kong Special Administration Region - 2012 by Architectural Services Department, Hong Kong.
 7. General Specification for Lift, Escalator and Passenger Coverage Installation in Government Buildings, Hong Kong - 2012 Edition by Architectural Services

Department, Hong Kong (hereafter referred to as Lift General Specification or Lift G.S.).

8. Design Manual - Barrier Free Access 2008 by Buildings Department, Hong Kong.
 9. Latest edition of British Standard 5655 by British Standard Institution, London.
 10. The Lifts and Escalators (Safety) Ordinance (Chapter 618) of the latest edition together with any amendments made thereto.
 11. EMSD Code of Practice on the Design and Construction of Lifts and Escalators of the latest edition with any amendment made thereto.
 12. EMSD Code of Practice for Lift Works and Escalator Works of the latest edition together with any amendment made thereto.
 13. EMSD Code of Practice for the Electricity (Wiring) Regulations of the latest edition together with any amendment made thereto, and BS 7671
 14. BD Code of Practice on Building Works for Lifts and Escalators of the latest edition together with any amendment made thereto.
 15. BD Building (Planning) Regulation of the latest edition together with any amendment made thereto.
 16. BD Code of Practice for Fire Safety in Buildings 2011 of the latest edition together with any amendment made thereto.
 17. The British Standard Specification BS EN81 and BS EN115 together with any amendment made thereto.
 18. Fire Services Department, Code of Practice (Minimum Fire Services Installations and Equipment and Inspection, Testing and Maintenance of Installations and Equipment), together with any amendment made thereto.
 19. Supply rules of the Power Supply Company.
 20. Any other relevant British Standard Specification.
 21. Code of Practice for Energy Efficiency of Building Services Installation by EMSD, 2012 together with any amendment made thereto.
 22. The British Standard Specification (PD 5304), Code of Practice for Safe Guarding for Machinery together with any amendment made thereto.
- B. The Contractor shall bear costs of all fees; permits, testing, certificates, etc., required by any of the above authorities for the operation of the lift.
- C. In the event of conflict between any of the provision of this Specification, this shall be referred to the Architect who shall decide as to which shall be followed.

1.04 ENVIRONMENTAL REQUIREMENTS

(NOT USED)

1.05 ENVIRONMENTAL, CLIMATIC & PSYCHOMETRIC CONDITIONS

(NOT USED)

1.06 SUBMITTALS

The Contractor shall submit the followings for the approval of Architect:-

- A. Installation detail drawings drawn to full scale of machine components and anchorage provision.

- B. All necessary drawings and documents for submission to The Electrical and Mechanical Services Department (EMSD) in accordance with EMSD requirement and to the satisfaction of Transport Department.
- C. All necessary drawings and structural justification for submission to the Buildings Department for approval.
- D. Sample material and performances data of equipment.
- E. Installation method

Comply with Manufacturer's detailed instructions in installing parking equipment. The actual sequence of installation of the works, the plants and machineries to be provided by the Contractor, and any temporary works to be done by the Contractor, as well as the erection sequence

F. Shop Drawings

- a) Shop drawings shall include instructions and explanatory for the sequence of installation of all materials. It is the responsibility of the Contractor to verify at the building site, all dimensions shown on Architect's drawings.
- b) No fabrication shall be commenced until the model is approved by the EMSD. Incorporate changes to shop drawings if required as result of modifications to the satisfaction of the EMSD.
- c) Approval of shop drawings shall not relieve the Contractor's duties and responsibilities under this Contract. No claim will be accepted for disapproved or amendments by the EMSD and, or Architect.

1.07 PRE-INSTALLATION CONFERENCE

(NOT USED)

1.08 DELIVERY STORAGE AND HANDLING

Package and store materials in a manner that will prevent damage, distortion, breakage or structural weakening. Replace any materials damage during manufacture, shipping, storage, or installation.

1.09 MOCK-UPS

This part shall be read in conjunction with Clause 8.03 of Specification Preliminaries for the requirements for prototypes and trial areas.

1.10 TESTING

A. Testing

The Contractor shall employ a Registered Lift Engineer under the Lifts and Escalators (Safety) Ordinance to undertake tests and examination of the complete installation. The tests and examination undertaken by the registered lift engineer shall include those specified in BS 5655: Part 1 and those recommended by the double deck car parking system manufacturer.

The relevant forms/certificates as required by the Lifts and Escalator (Safety) Ordinance shall be signed and submitted by the registered lift engineer to the Architect on completion of the double deck car parking system installation.

A test record containing information as required in Appendix D of BS 5655: Part I either in the sequence as shown in the appendix or in a form as recommended by the manufacturer and approved by the Architect shall be submitted.

Tests which purely demonstrate the performance characteristic of the lift shall be performed in the presence of the Architect at the acceptance of the installation.

At the end of the free maintenance period a further test shall be carried out by the Contractor and any defects found shall be rectified by the Contractor without charge to the Employer.

The format of testing & commissioning procedures and testing report in hard and soft copies are as follows:-

- a) copies for approval
- b) copies for distribution after approval

B. O&M Manuals

1. One (1) month prior to Substantial Completion, the Contractor shall submit to the Architect for approval three nos. of draft copies of O & M Manual containing-full technical description of the double deck car parking system, equipment schedules, catalogues, manufacturers' operating instructions and manuals, parts lists, list of record drawings, list of test certificates and detailed maintenance schedules.
2. After approval by the Architect and within one month from Substantial Completion, the Contractor shall submit final O & M Manuals bound with hard covers to the Architect for record.

1.11 WARRANTY

- A. This part shall be read in conjunction with Clause 8.04 of Specification Preliminaries for the list of material/ system requiring joint written guarantees.
- B. In addition to guarantee submission from the manufacturers suppliers as required under the other specification clauses, submit write guarantee, signed by the Contractor for the whole Double Deck Parking System including design, material, finishes and workmanship to satisfy the performance requirements as specified for, and agree to repair or replace defective materials and workmanship and during the guarantee period. The Contractor shall be responsible for damages to building and furnishings caused by defective materials and workmanship during the guarantee period.
- C. Defective materials and workmanship hereby decided to include evidence of abnormal deterioration, loss of mechanical properties, structural failure, corrosion, distortion, aging or weathering of the work, structural failure of components, failure of operation parts to function normally, deterioration of discoloration of finishes of normal weathering aging, and failure of the work to fulfill other specified performance requirements.
- D. Repairs or replacements during the guarantee period required unforeseen Circumstances (which exceed performance requirements), alterations, abuses of the work, vandalism, failure of the supporting structure and other causes beyond the Contract.
- E. This guarantee and its enforcement shall not deprive the Employer of other action, right or remedy available to him.

1.12 SPARE PARTS

This part shall be read in conjunction with Clause 8.05 of Specification Preliminaries for the list of spares and spare parts.

PART 2 PRODUCT

2.01 APPROVED MANUFACTURER

Subject to compliance with the Project Contract Documents, provide panel ceilings and required accessories from one or more of the following manufacturers, or equivalent approved by the Architect:

A. Arrowtech Engineering (H.K.) Co., Ltd.

B7, 9/F., Merit Industrial Centre,

No. 94 To Kwa Wan Road,

Kowloon, Hong Kong

Tel: (852) 2364 0727

Email: alex.to@arrowtech.com.hk

B. Equivalent approved by the Architect

2.02 MATERIALS / SYSTEMS

The Double Deck Parking System shall include the followings:-

A. Control Panel

Each mechanical car parking system shall be manual operated and have a key and push button with a key locked switch for operation of the machine. Adequate safety and control features shall be provided, but not limited to the following, to the satisfaction of EMSD/ any other authority having jurisdiction over the installation: -

- a. Safe and adequate access to the platform controls shall be provided for the operator / user.
- b. Photo-Electric Switch Devices
Installed at the entrance to each parking space, the photo- electric system will automatically shut down all operations if a car fails to come to a complete stop, is too long for the parking space, or if some foreign object impairing its movement. Operations can only be resumed after the system is reset, thus ensuring maximum safety when cars are being parked.
- c. Warning Devices
A series of warning devices enhance safety by flashing an alert light or sounding a warning device whenever the operator or a pedestrian comes too close.
- d. Emergency Stop Devices
Installed on system control panel, emergency stop buttons can be used to shut down the system completely in case of any unforeseen emergency.
- e. Residue Current Devices
This safety feature is designed to prevent electrical leakage in wet and humid conditions.
- f. Overload Current Devices
This safety feature is designed to prevent electrical overload of the driving parts or unexpected locking of the parking system.

g. Up / Down Positioning Devices

This feature automatically suspends system operations whenever elevator platforms are moving cars up or down to their assigned spaces.

h. Fall Prevention Devices

The fall prevention feature uses safety hooks for prevention of damage if hydraulic pressure in the elevator platforms fails.

i. Notice in Chinese and English stipulating the operation instruction, the safe working load and maximum allowable vehicle size shall be provided.

B. Safety Device

Adequate safety devices shall be designed and provided, but not limited to the following, to the satisfaction of EMSD / any other authority having jurisdiction over the installation: -

1. Mechanical Locks

The platform holding the motor vehicle shall be locked automatically when it reaches the upper stop position. A visual indication or audible signal will be given, if the platform has not reached the upper or lower stop position or has not been mechanically locked.

2. Overload Device

Unless there are other means to prevent lifting a motor vehicle exceeding the capacity of the platform / equipment, an overload device should be installed to prevent any movement of the platform and to give an alarm in case of the load on the elevating platform is in excess of the rated capacity.

3. Deadman Control

The movement of the platform shall be stopped, unless the operator keeps his hand on the local control device during the raising or lowering process.

4. Key Switch

A key-locked switch to avoid any unauthorized operation shall be provided.

5. Chocks

Automatic or permanent chocks shall be provided to arrest any unpowered movement of the vehicle on the platform.

6. f) Emergency Operation

A mechanical device shall be provided to raise and lower the platform when the normal power supply fails.

7. g) Other safety devices shall include but not limited to the following:-

- a. Entrance eye-beam protective devices
- b. Limit switch for prevention of pallet overrun
- c. Hooks for anti-pallet falling
- d. One phase and reverse phase protective devices
- e. Motor and operation brake
- f. Vehicle positioning eye-beam detector
- g. Broken chain detector
- h. Prevention of driving chain displacement

PART 3 EXECUTION

3.01 EXAMINATION / INSPECTION

- A. The driving system and supporting castors shall be visually inspected before erection of cover plates.
- B. The system shall be functional tested with the specified loading capacity after completion.

3.02 PREPARATION

(NOT USED)

3.03 INSTALLATION / APPLICATION

- A. General
 - 1. Examine condition under which parking systems are to be installed for compliance with requirements for installation tolerances, critical dimensions and other condition affection performance.
 - 2. Comply with manufacturer's detailed instructions when installing parking equipment.
 - 3. Examine roughing-in for electrical systems to verify actual locations of connections before parking machine installation.
- B. Adjustment, Cleaning, Finishing and Protection
 - 1. Make necessary adjustments for safe and efficient operation.
 - 2. Clean exposed and semi-exposed surfaces. Touch-up shop- applied finishes to restore damaged or soiled area.
- C. Sound Reducing and Vibration Elimination
 - 1. All equipment shall be quiet in operation, free of cracking or roaring noise. Sound reducing rubber pads, shock absorption buffers, acoustic fibre glass lining or other effective means shall be provided where necessary to eliminate vibration and noise transmission. The Contractor shall provide all additional measures as required so that the noise created during the operation of all equipment and machinery will be reduced to a level acceptable to the Architect.
 - 2. The Contract price shall deem to have included for comprehensive control of noise and vibration from all mechanical parking drive equipment and switching, car movement, trips breaking, signaling. Noise levels during operation of lifts shall not exceed 65 dB(A) measured at 1m from any one of the noise generating sources and vibration isolation for drive units and switching must be sufficient to ensure no measurable increase in noise levels in local occupied areas during operation of the mechanical parking system. In addition, the vibration levels shall not exceed the prescribed criteria given in ISO 2631-2.
- D. The Contractor shall be responsible for the installation of the equipment such that it shall operate in a balanced and aligned condition. The equipment shall conform, not greater than the range of vibration severity recommended by ISO 10816.
- E. During operation the horizontal and vertical vibration, measured on the car platform, shall not exceed 0.1m/s² peak to peak, and 1.0m/s² respectively.
- F. Detailed proposals for noise and vibration control shall be submitted for approval.

G. Electrical Supply and Electrical Installation

1. The contractor shall provide power supply cable complete with an isolating switch adjacent to each set of double deck car parking mechanical system.
2. The contractor shall provide all electrical connections from the isolating switch, including the outgoing wiring either in trunking or conduits / the sub-main switches / distribution boards to the machines, control equipment, lighting and power, earthing bonding etc. for the mechanical parking system. All the electrical installation by mechanical parking Contractor shall be weatherproof type except those approved by architect / engineer.
3. Electricity supply shall be 380 volts/3 phase/50 Hertz or 220 volts/single phase/50 Hertz.
4. In the event of failure of electricity supply or fire breaks out, a mechanical device shall be provided to raise and lower the platform by the Contractor.
5. All exposed metal work liable to become electrically charged shall be efficiently bonded and earthed. All wiring shall be enclosed in galvanized steel conduits or trunking. Steel conduit, except flexible conduits shall be of heavy gauge and screwed.

3.04 TOLERANCES

(NOT USED)

3.05 CLEANING

(NOT USED)

3.06 PROTECTION

(NOT USED)

3.07 MAINTENANCE

A. Free Maintenance

The Contractor shall furnish maintenance free of charge for the entire installation for a period of twelve months following the date of Substantial Completion. The extent of work required to be carried out should cover at least, but not limited to the following:

1. Planned Maintenance
 - a. To be responsible for any repairs necessary to maintain the installation in good and safe working order at all times.
 - b. To dispatch competent and specially trained workmen once weekly during normal working hours to maintain each double deck car parking system in accordance with the General Specification for Lift, Escalator and Passenger Coverage Installation in Government Buildings, Hong Kong - 2012 Edition by Architectural Services Department, Hong Kong (hereafter referred to as Lift G.S.), Annex IA Maintenance Schedule for Electrical Passenger and Goods Lifts, where applicable.
 - c. To supply all lubricants, cleaning materials, rope preservatives etc. required.
 - d. Replace all burnt out lamp bulbs/tubes with bulbs/tubes of correct rating.
 - e. To provide, repair or replace at no additional cost to the Employer such mechanical and electrical parts of the installation required for the safe and efficient operation of the installation.

2. Emergency Maintenance
 - a. To provide a "call-out" service during and outside normal working hours to carry out emergency maintenance by competent workman.
 - b. To attend to any breakdowns reported to him by telephone, (or other means) with the utmost effort and in no circumstance attendance to the breakdown shall exceed 30 minutes after the call is received.
3. Statutory Maintenance
4. To carry out periodic testing and examination of safety equipment as may be required by the provisions of any enactment in force relating thereto or of any enactment, regulation or by-law of any local or other duly constituted authority which may be applicable to such tests and to provide such copies of the test certificate, duly signed by a Registered Lift Engineer, as may be required. Provided always that any renewals or repairs necessitated by reason of negligence or misuse of the equipment or by reason of any other cause beyond the Contractor's control with the exception of wear and tear shall be carried out by the Contractor if so required by the Employer at an additional cost to be negotiated by both parties.

All work under this maintenance provision shall be performed by competent personnel under the supervision and in the direct employ of the Contractor.

B. Fee Proposal for Maintenance after Free Maintenance Period

1. The tenderer shall submit with his tender a fee proposal for undertaking maintenance of the installation provided under the Contract for a period of five (5) years from the date of expiration of the Free Maintenance Period. This maintenance proposal shall be prepared on a fully comprehensive maintenance basis including all necessary materials and skilled labours.

C. Training

1. The Contractor shall provide adequate training to employer's personnel regarding operation and maintenance of the installation.
2. Complete, printed or typed instruction booklets shall be provided covering maintenance, operation and adjustment of each piece of equipment.
3. Training shall cover at least, but not be limited to the following:
 - (i) familiarization with the equipment supplied;
 - (ii) technical details with regard to control principles and techniques employed;
 - (iii) installation and alignment technique;
 - (iv) testing and adjusting procedures;
 - (v) routine maintenance and overhaul.

END OF SECTION